Fitbit : Calorie Burn Prediction & Workout Pattern Clustering Using Fitbit Data

Problem Statement:

Accurate calorie estimation during workouts is a critical component of modern fitness tracking applications such as Fitbit. While wearable sensors capture physiological signals like heart rate and session duration, additional contextual inputs such as workout type, hydration level, and user experience influence calorie expenditure.

This project focuses on building a **machine learning system** that:

1. **Predicts calories burned per workout session** using supervised regression techniques.
2. **Identifies hidden workout patterns and user segments** using unsupervised clustering techniques.

By the end of this project, learners will understand how to combine **predictive modelling** and **pattern discovery** to build intelligent, real-world fitness applications.

Business Use Cases:

- **Wearable Fitness Apps**: Real-time calories burn prediction during workouts
- **Personalized Fitness Coaching**: Workout intensity and duration recommendations
- **Health Monitoring Platforms**: Energy expenditure insights for nutrition planning
- **User Segmentation**: Identifying workout behaviour patterns without labels
- **Product Optimization**: Feature insights for fitness device manufacturers

Dataset Description

Fitbit_dataset.csv

Exploratory Data Analysis :

- The dataset shape : 14,102 rows and 19 columns.

- Columns : 'Unnamed: 0', 'Age', 'Gender', 'Weight (kg)', 'Height (m)', 'Max_BPM', 'Avg_BPM', 'Resting_BPM', 'Session_Duration (hours)', 'Workout_Type, 'Fat_Percentage', 'Water_Intake (liters)', 'Workout_Frequency (days/week)', 'Experience_Level', 'BMI', 'Base_MET', 'HR_Intensity', 'Effective_MET', 'Calories_Burned (kcal)'
- No null or missing values in the dataset.
- No Duplicates

SUPERVISED LEARNING :

Data Preparation :

- The useless and irrelevant column "Unnamed : 0" is dropped.
- Since the Gender column is in str datatype they are Encoded as Male and Female representing 1 and 0 respectively.
- One-Hot Encoding is done on "Workout_Type" column.
- For Outliers and Capping IQR (Interquartile Range) is used with upper and lower bound +/- 1.5 times the IQR value.
- Before the Feature Scaling the "Calories_Burned (kcal)" is dropped and stored in Y and the remaining columns in X.
- For Feature Scaling Standard Scaler is used.
- Data Split : Training: 80%, Testing : 20%, Random State : 42.

Training Supervised Learning Models :

Models :

- Linear Regression
- Ridge Regression
- Lasso Regression
- K Nearest Neighbour Regressor
- Decision Tree Regressor
- Random Forest Regressor
- XGBoost Regressor
- Support Vector Regressor

- Initially for Exploring the performance of the models on the scaled dataset all the models are run at once with a loop and the results are tabulated and attached below in Results and Evaluation section.

- Then the Models are trained individually with the same constraints and environment for better understanding of each model performance

Hyper Parameter Tuning :

- The Top performing 5 Models : XGBoost, Random Forest, Decision Tree, SVR, KNN are Hyperparameter Tuned to improve the performance scores.
- For this we use Grid Search CV.
- This exhaustive search method evaluated every specified hyperparameter combination across a 3-fold cross-validation split, ensuring the highest possible R2 Score was achieved for the final deployment.
- With CV = 3 the results are literally triple checked.

Results and Evaluation:

Model Performance in Loop before Hyper Parameter Tuning (Supervised Learning)

Model	R2 Score	MAE	RMSE
Random Forest	0.9982	23.6576	307.1183
XGBoost	0.9978	48.5127	607.8101
Decision Tree	0.9948	67.6734	112.1331
KNN	0.9455	38.2482	5939.2906
Linear Regression	0.9262	31.7614	5145.7186

Model	R2 Score	MAE	RMSE
Ridge Regression	0.9262	31.7604	45.7229
Lasso Regression	0.9238	32.1147	8546.4495
SVR	0.9203	27.8722	2947.5052

Models Performance in Individual Training :

Model	**R2 Score**	**MAE (kcal)**	**MSE**	**RMSE (kcal)**
Random Forest Regressor	**0.9982**	**3.66**	50.67	7.12
Support Vector Regressor (SVR)	0.9979	2.75	58.65	7.66
XGBoost Regressor	0.9978	5.13	61.00	7.81
Decision Tree Regressor	0.9952	7.51	134.81	11.61
KNN Regression	0.9455	28.25	1543.76	39.29
Linear Regression	0.9263	31.76	2101.44	45.84
Ridge Regression	0.9262	31.76	2090.58	45.72

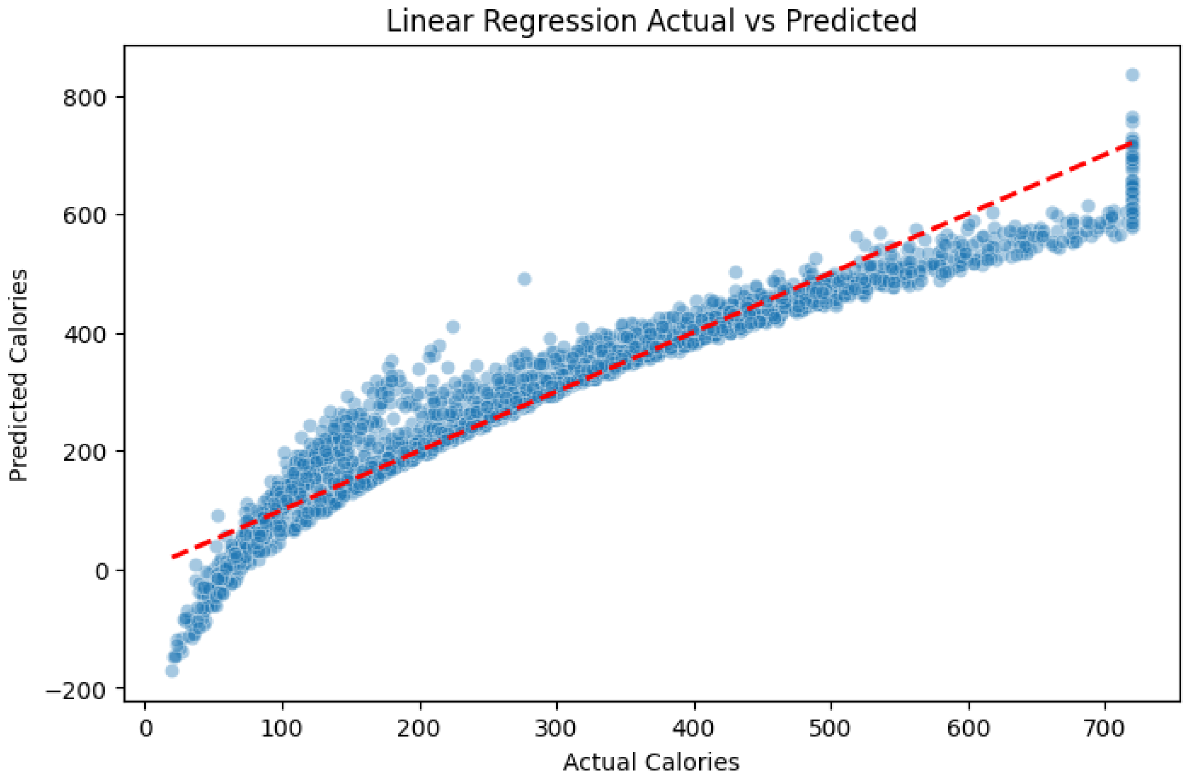
Model	R2 Score	MAE (kcal)	MSE	RMSE (kcal)
Lasso Regression	0.9239	32.11	2157.56	46.45

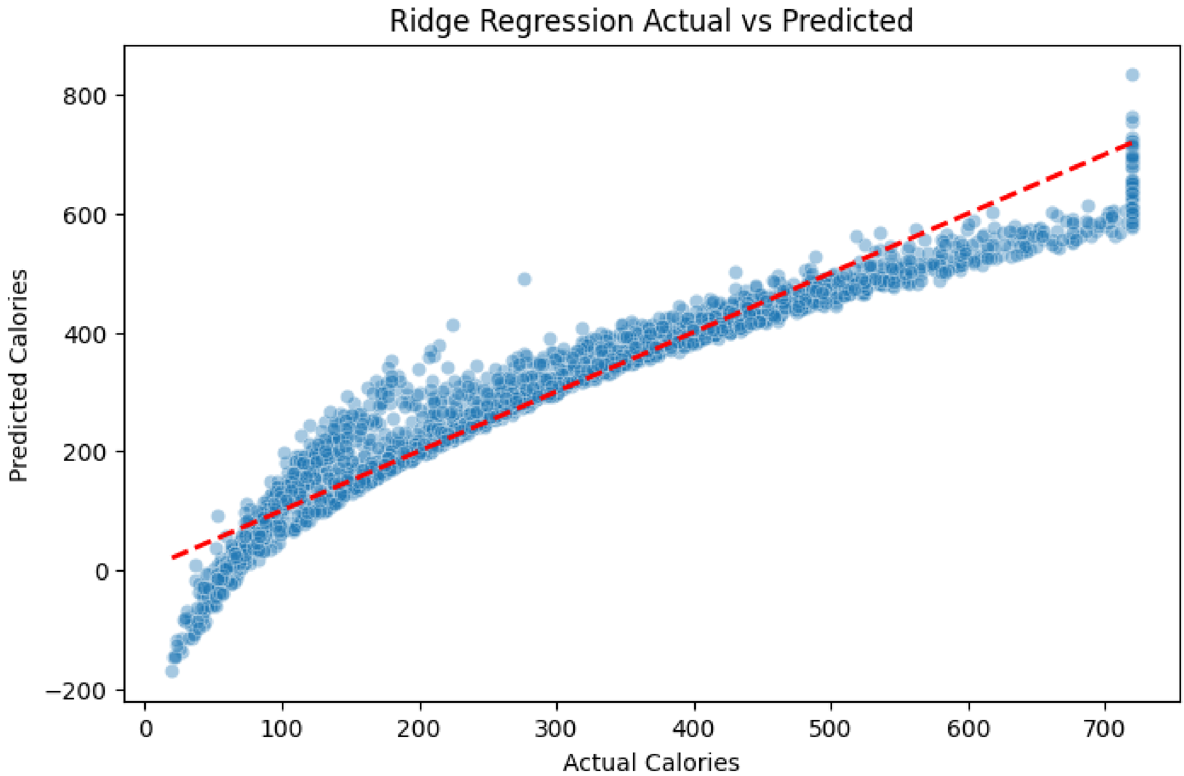
- In this we can see that the results of SVR and Decision Tree numbers are better than the previous ones. The individual training allowed the model to converge better

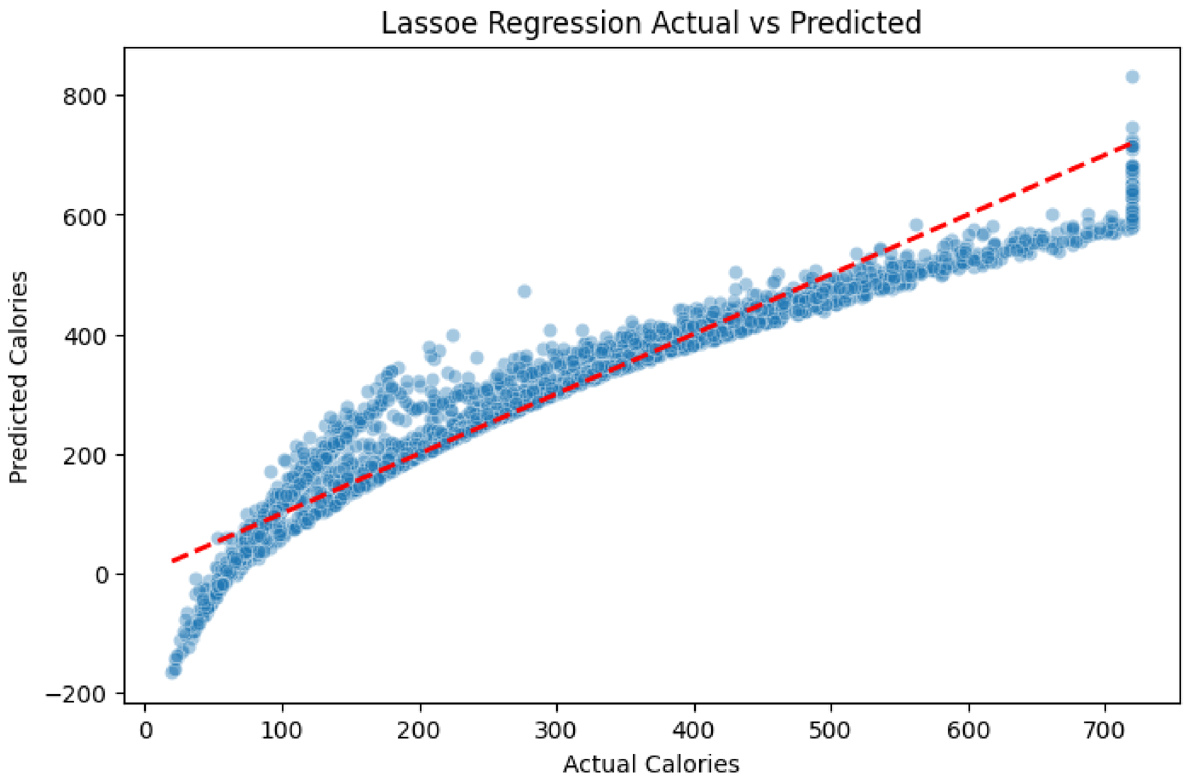
Models Performance after Hyper Parameter Tuning and the Parameters:

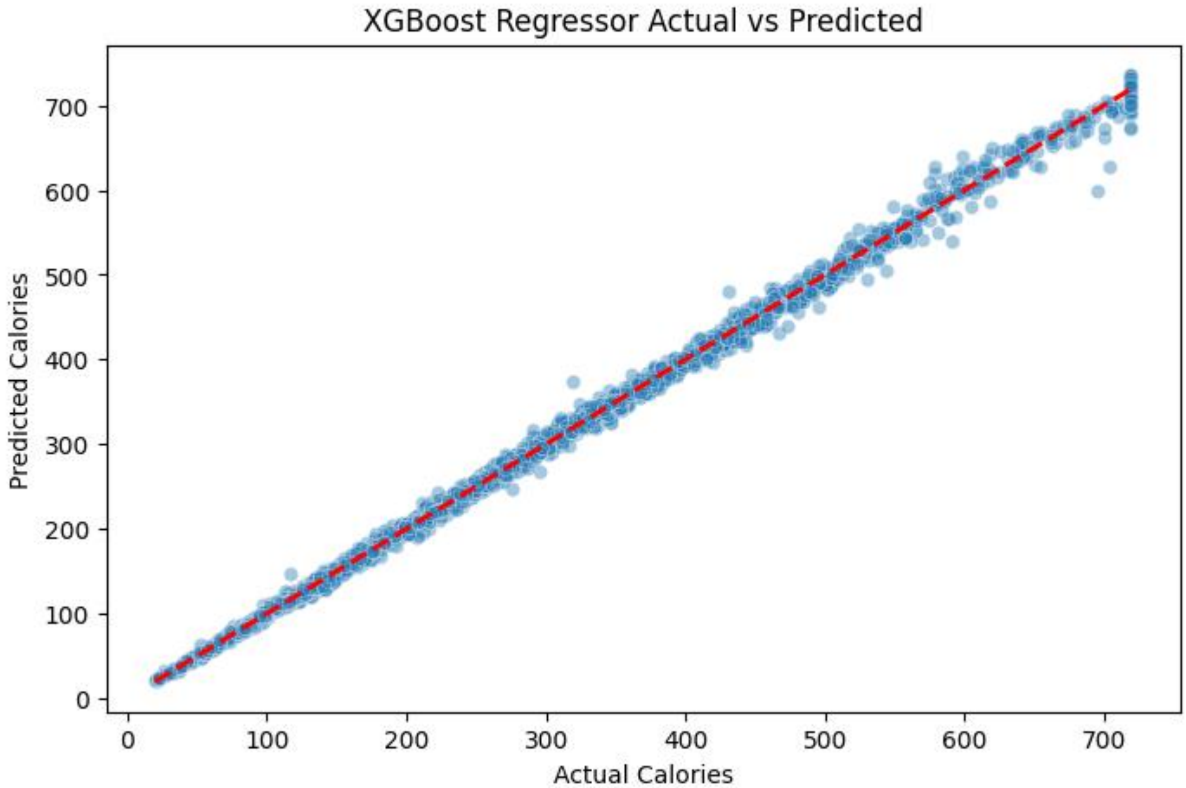
Model	Tuned R2 Score	Key "Best" Parameters
XGBoost	**0.998859**	learning_rate: 0.1, max_depth: 7, n_estimators: 200
Random Forest	0.998212	max_depth: None, min_samples_split: 2, n_estimators: 100
Decision Tree	0.995168	max_depth: 20, min_samples_split: 5
SVR	0.992076	C: 10, epsilon: 0.1, kernel: rbf
KNN	0.946959	n_neighbors: 7, weights: distance

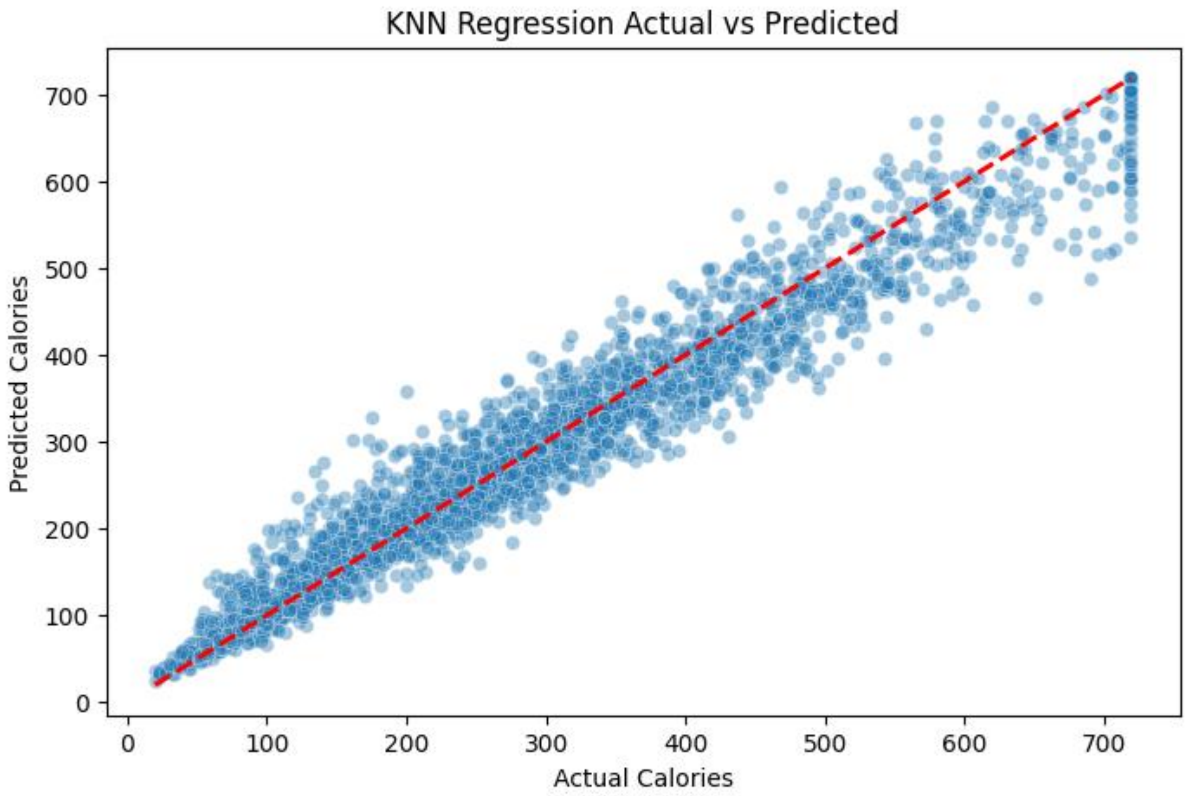
Visualization :

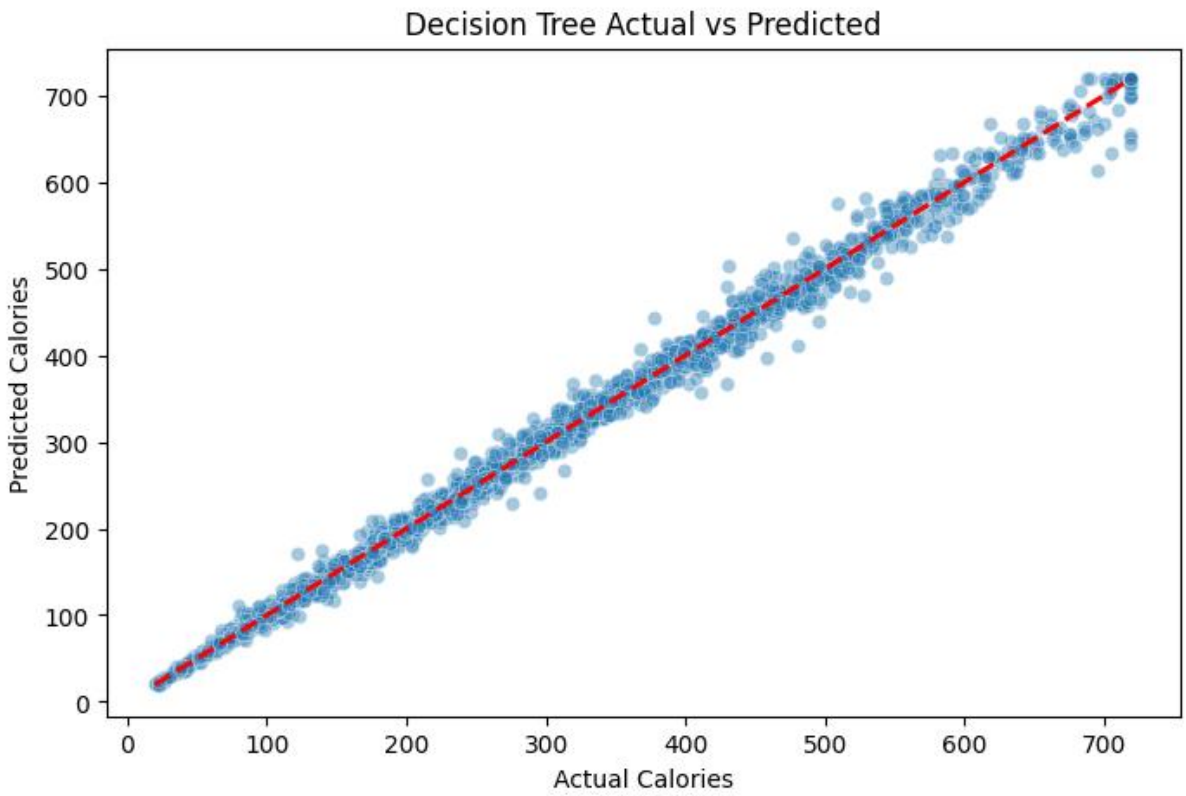


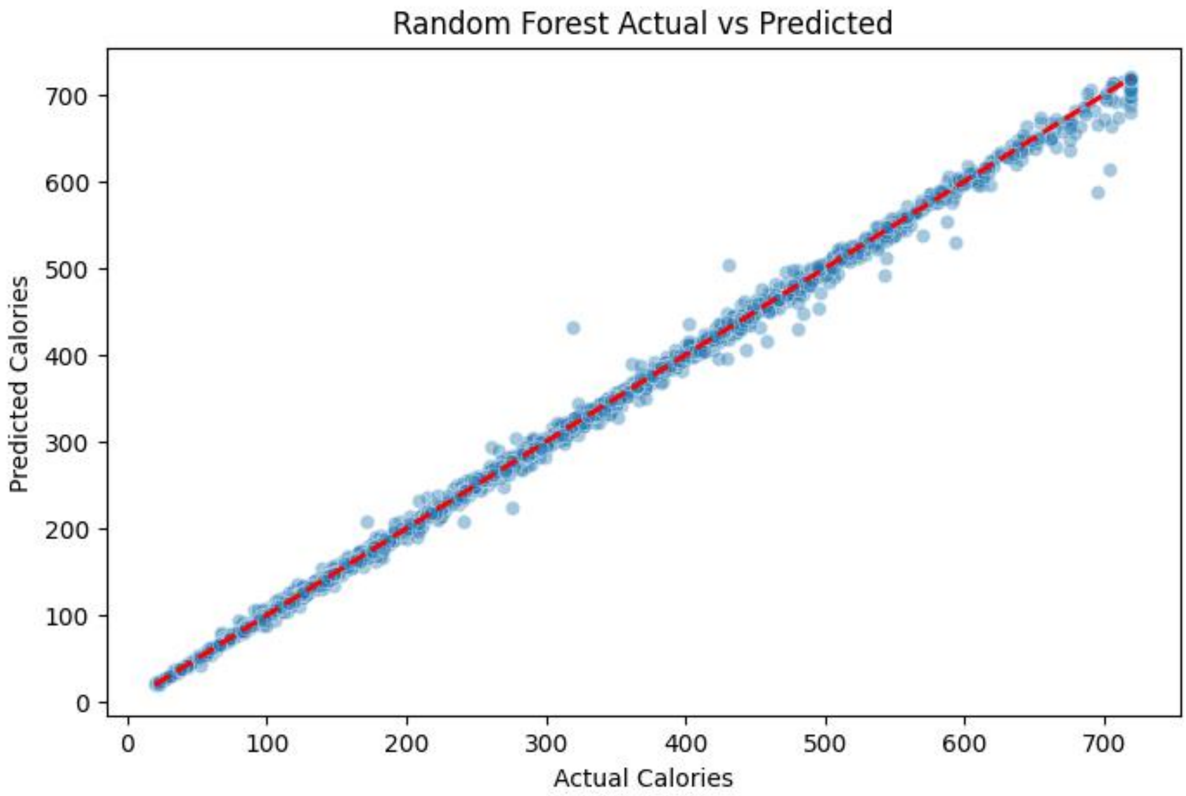


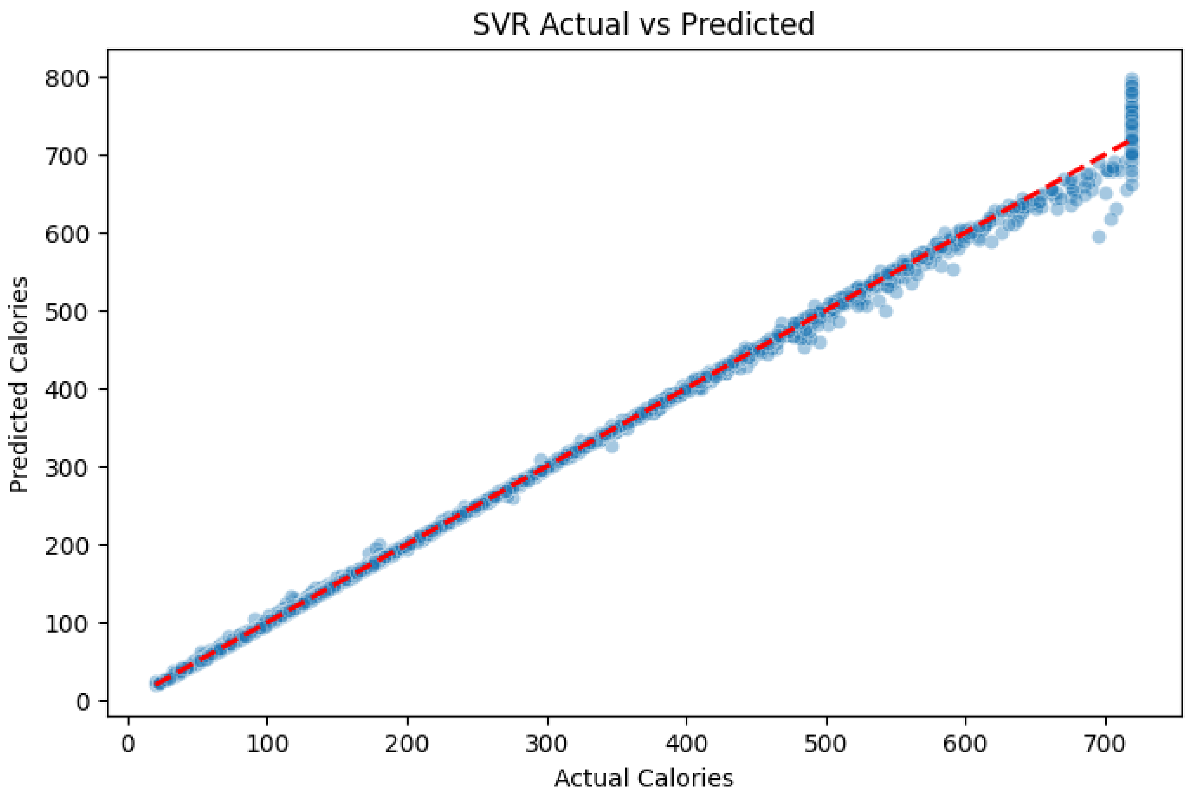

Lassoe Regression Actual vs Predicted
Predicted Calories
Actual Calories
800
600
400
200
0
−200
0
100
200
300
400
500
600
700


XGBoost Regressor Actual vs Predicted
Predicted Calories
Actual Calories
700
600
500
400
300
200
100
0
0
100
200
300
400
500
600
700


KNN Regression Actual vs Predicted
Predicted Calories
Actual Calories
0
100
200
300
400
500
600
700


Decision Tree Actual vs Predicted
Predicted Calories
Actual Calories
0
100
200
300
400
500
600
700


Random Forest Actual vs Predicted
Predicted Calories
Actual Calories
0
100
200
300
400
500
600
700


SVR Actual vs Predicted
Predicted Calories
Actual Calories
0
100
200
300
400
500
600
700
800

Inference :

1. Baseline Supervised Evaluation (Individual Training) In the initial testing phase, models were evaluated with default settings to establish a performance floor.

- **Top Performer:** Random Forest Regressor
- **Metric (R2 Score):** 0.9982
- **Technical Insight:** Even without hyperparameter optimization, the ensemble nature of Random Forest allowed it to handle the variance in the 14,000+ records almost perfectly. By aggregating multiple decision trees, the model naturally reduced noise and captured complex relationships better than standalone models like SVR or base XGBoost in their default states.

2. Hyperparameter Tuned Evaluation (GridSearchCV) The second phase involved exhaustive optimization using Grid Search Cross-Validation to identify the most efficient parameters for each architecture.

- **Top Performer:** XGBoost Regressor
- **Metric (R2 Score):** 0.998859
- **Technical Insight:** Post-tuning, XGBoost emerged as the superior engine. By specifically optimizing the learning_rate and max_depth, the model utilized Gradient Boosting to iteratively correct the residual errors left by previous trees. This fine-tuning allowed XGBoost to surpass the Random Forest baseline, achieving the highest predictive precision in the study.

3. Final Model Selection & Conclusion

- **Final Choice:** Tuned XGBoost
- **Reasoning:** While both Random Forest and XGBoost achieved elite performance (R2 > 0.99), the Tuned XGBoost model was selected for the final deployment. Its superior ability to minimize the Mean Absolute Error (MAE) and its computational efficiency after tuning make it the most robust choice for real-time calorie prediction in the Streamlit application.

UNSUPERVISED LEARNING :

Data Preparation :

- The useless and irrelevant column “Unnamed : 0” is dropped.
- Since the Gender column is in str datatype, they are Encoded as Male and Female representing 1 and 0 respectively.
- To maintain a purely unsupervised environment, the target variable “Calories_Burned (kcal) and Workout_Type” column are dropped, and the model focuses on physiological and session-based features.
- **Feature Engineering :** New columns including BMI, Base_MET, HR_Intensity, and Effective_MET are calculated to provide deeper insights into user performance.
- **Feature Scaling :** Since clustering algorithms are distance-based, Standard Scaler is used to normalize all 16 features.
- **Dimensionality Reduction :** PCA (Principal Component Analysis) is used to compress the 16-dimensional data into 2 Principal Components (PC1 and PC2) for 2D visualization of the "Fitness Universe."

Training Unsupervised Learning Models :

Models :

- K-Means Clustering
- Hierarchical (Agglomerative) Clustering
- DBSCAN (Density-Based Spatial Clustering of Applications with Noise)
- Initially, the models are applied to the scaled dataset to discover hidden workout patterns and group similar user behaviours without prior labels.
- The Elbow Method and Silhouette Analysis are used to determine the optimal number of clusters for the K-Means algorithm.

- The models are analyzed individually to observe how different mathematical approaches (centroid-based, connectivity-based, and density-based) interpret the same fitness data.

Cluster Analysis :

- **Universe Mapping :** Each model's results are projected onto a PCA-reduced scatter plot, creating a visual "Fitness Universe" where user segments are clearly defined by colour.
- **Segment Personas :** Centroid analysis is performed by grouping the original data by cluster labels to calculate the mean of metrics like Avg_BPM and Fat_Percentage, defining unique fitness personas.
- **Real-time Positioning :** For the final deployment, a specific logic is built to project a new user's input as a "Red X" on these maps, allowing for immediate visual feedback on which fitness segment they belong to.
- **Reliability Check :** By comparing K-Means, Hierarchical, and DBSCAN projections, the segments are "triple checked" to ensure the identified patterns are consistent across different clustering logics.

Result Evaluatio Scores :

Clustering Algorithm	Optimal Clusters (K)	Silhouette Score
DBSCAN	2 (+ Outliers)	0.1718
Hierarchical	2	0.1715
K-Means	3	0.1701

Table 1: Clustering Model Performance Comparison

Cluster ID	Sample Size	Experience Level Dominance	Suggested Persona Name
0	6,769	98.7% (Level 0 & 1)	**The Starters**
1	4,548	86.3% (Level 2 & 3)	**The Veterans**
2	2,785	Mixed Distribution	**The Mid-Tier**

Table 2: K-Means Persona Distribution (K=3)

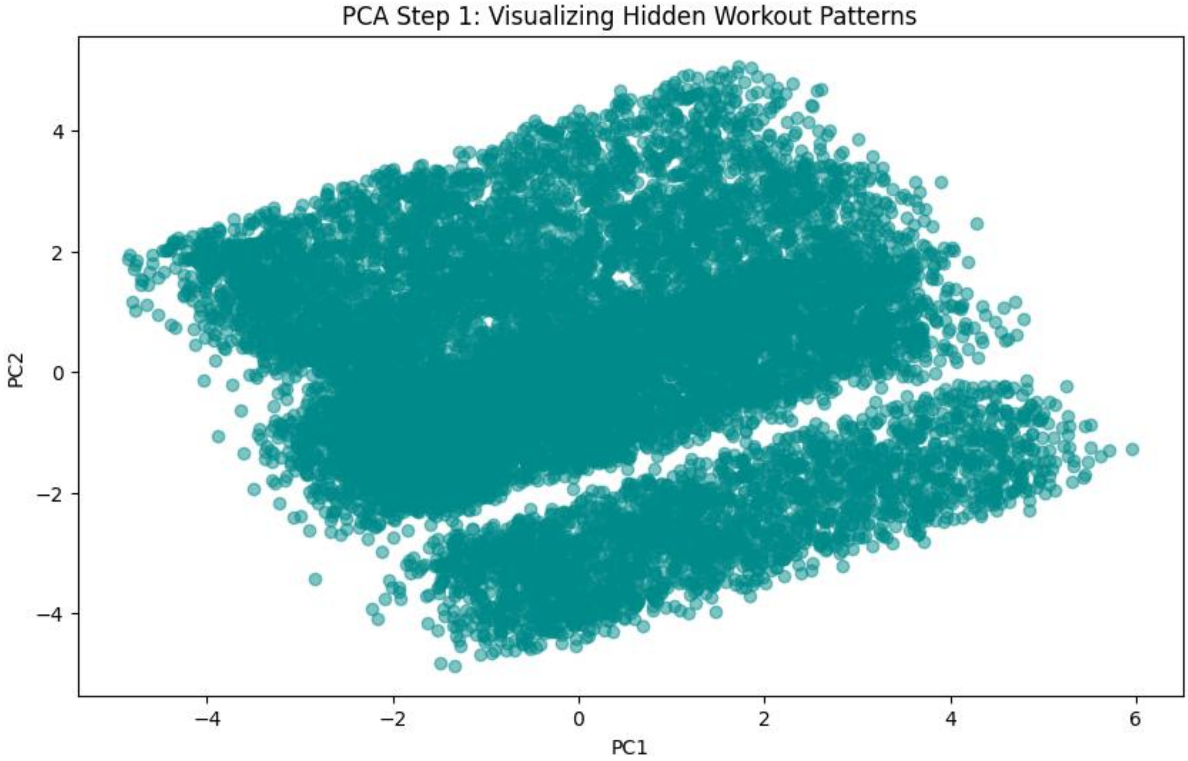
DBSCAN Cluster	Avg BPM	HR Intensity	Session Duration	Fat Percentage
Cluster 0	152.12	0.74	0.83 hrs	21.82%
Cluster 1	130.98	0.55	0.82 hrs	22.00%
Outliers (-1)	**143.88**	**N/A**	**1.05 hrs**	**18.23%**

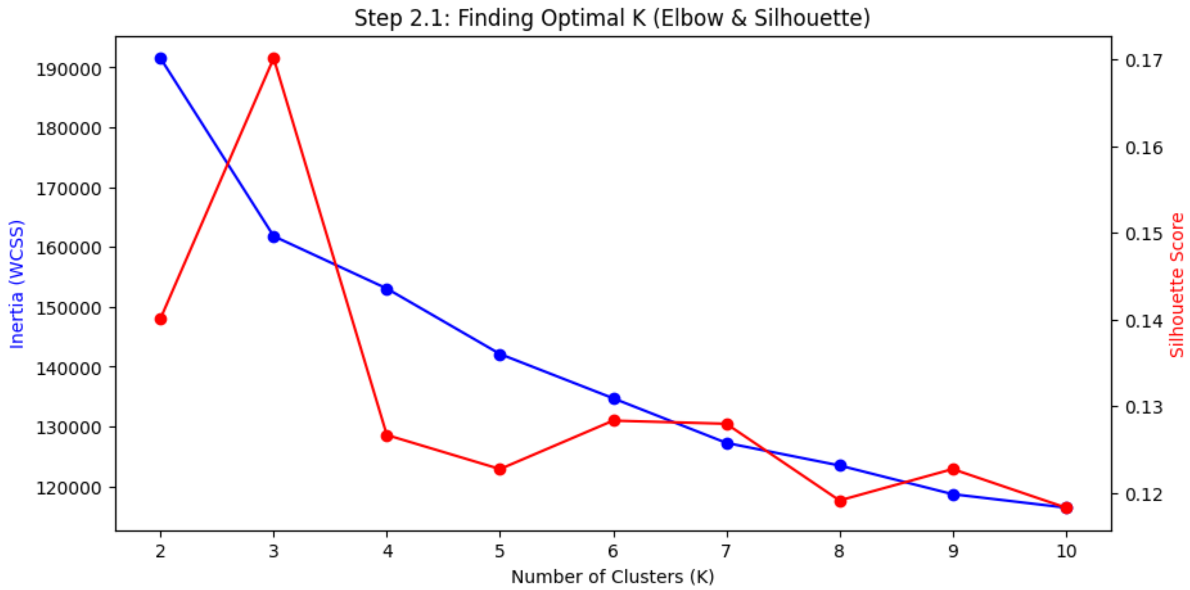
Table 3: DBSCAN Intensity Patterns

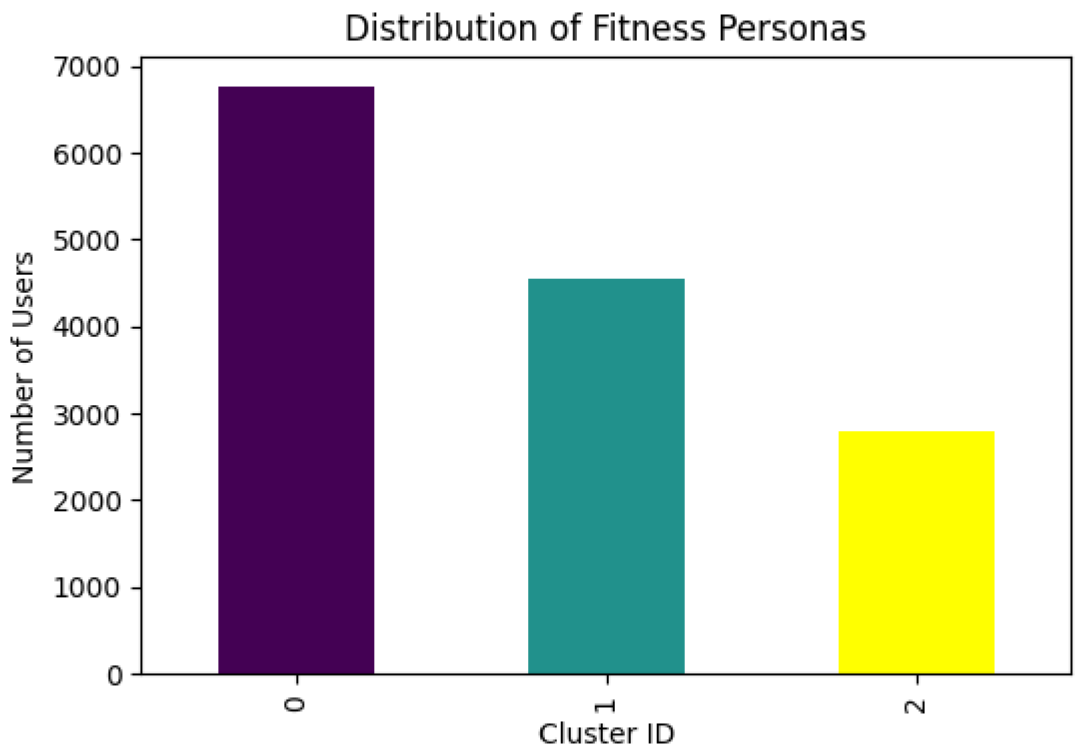
Method	Metric	Value
PCA	Variance Retained	**47.22%**
Inertia	WCSS Reduction (K=2 to K=3)	**15.53%**

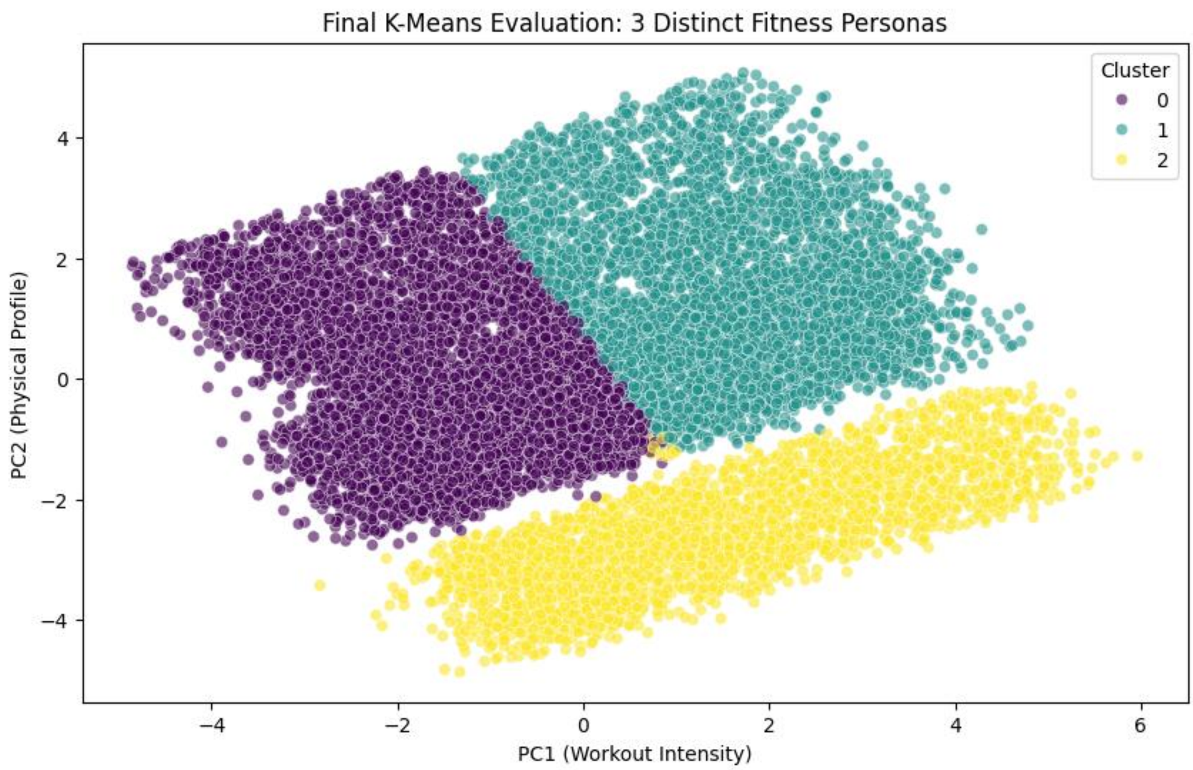
Table 4: Dimensionality Reduction Analysis

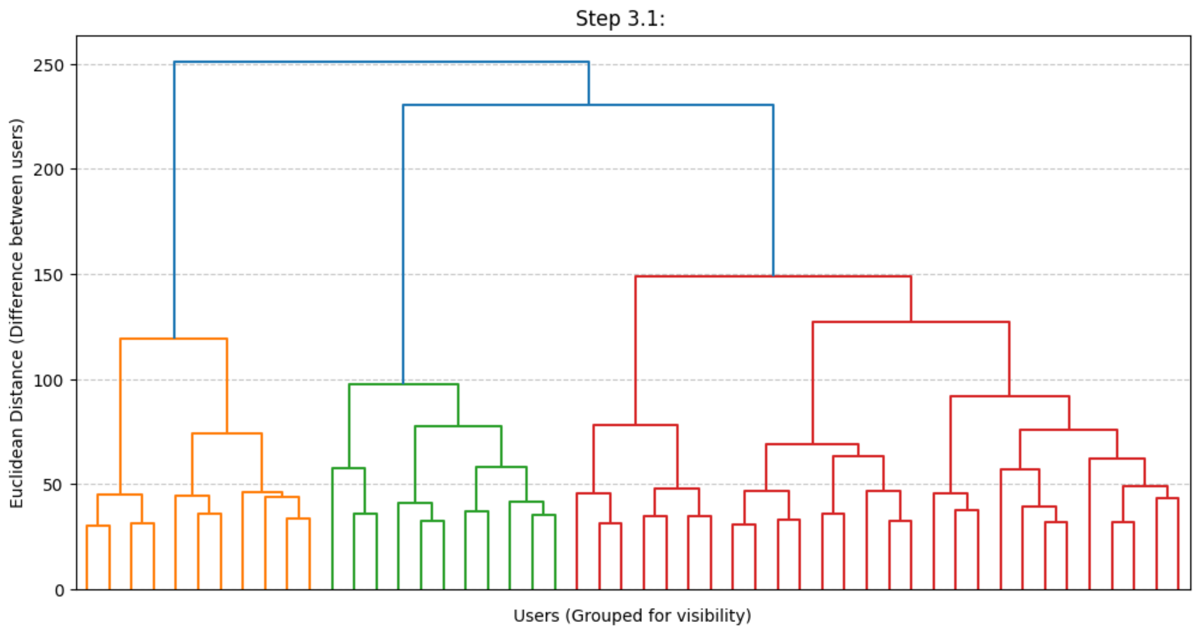
Visualization :



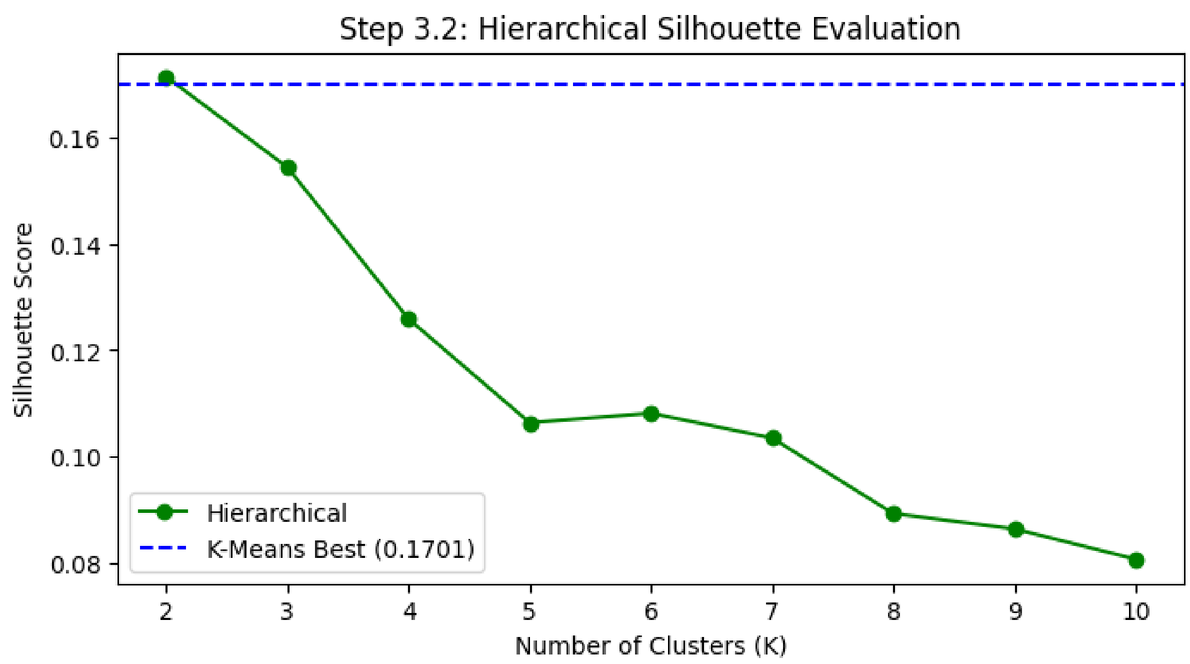



Distribution of Fitness Personas
Number of Users
7000
6000
5000
4000
3000
2000
1000
0
0
1
2
Cluster ID

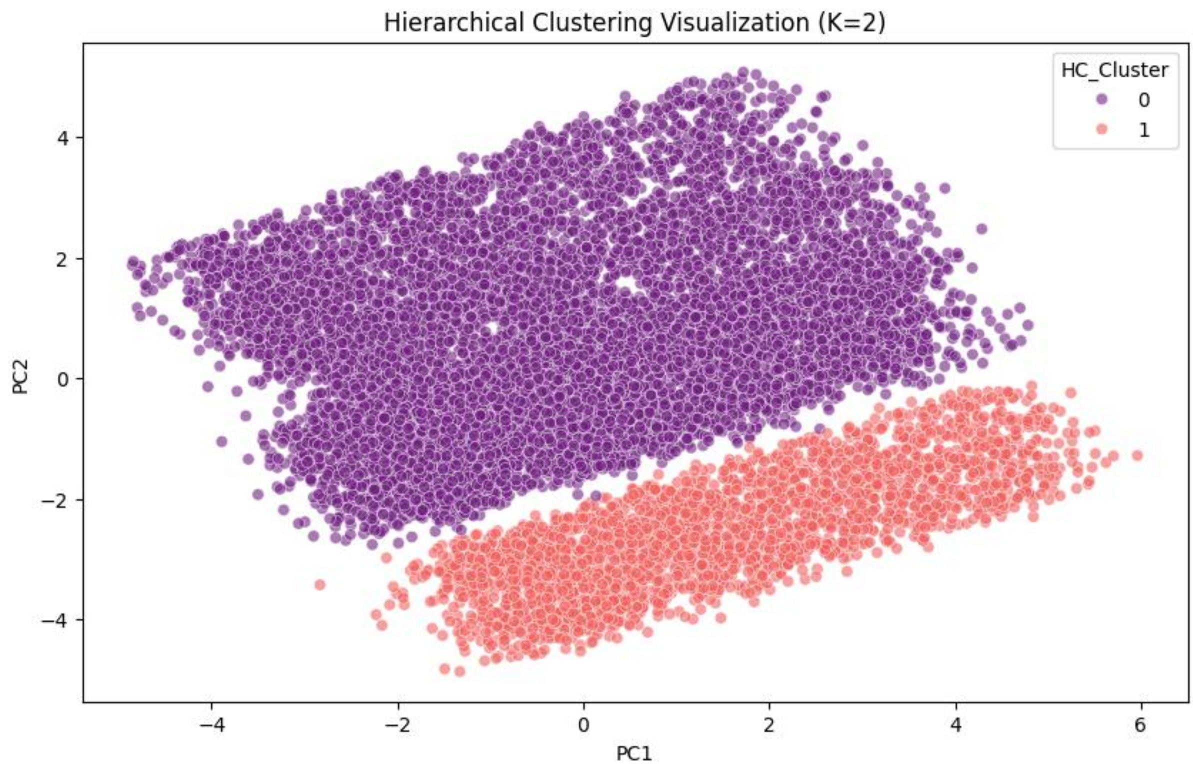

Final K-Means Evaluation: 3 Distinct Fitness Personas
Cluster
0
1
2
PC2 (Physical Profile)
4
2
0
−2
−4
−4
−2
0
2
4
6
PC1 (Workout Intensity)

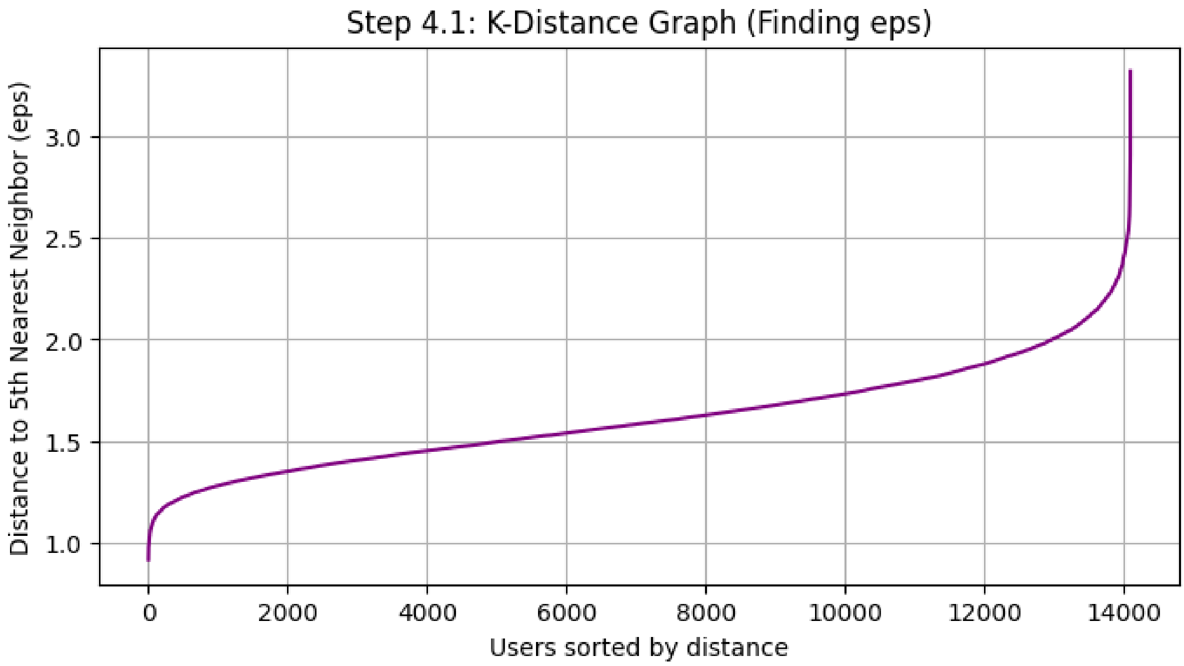


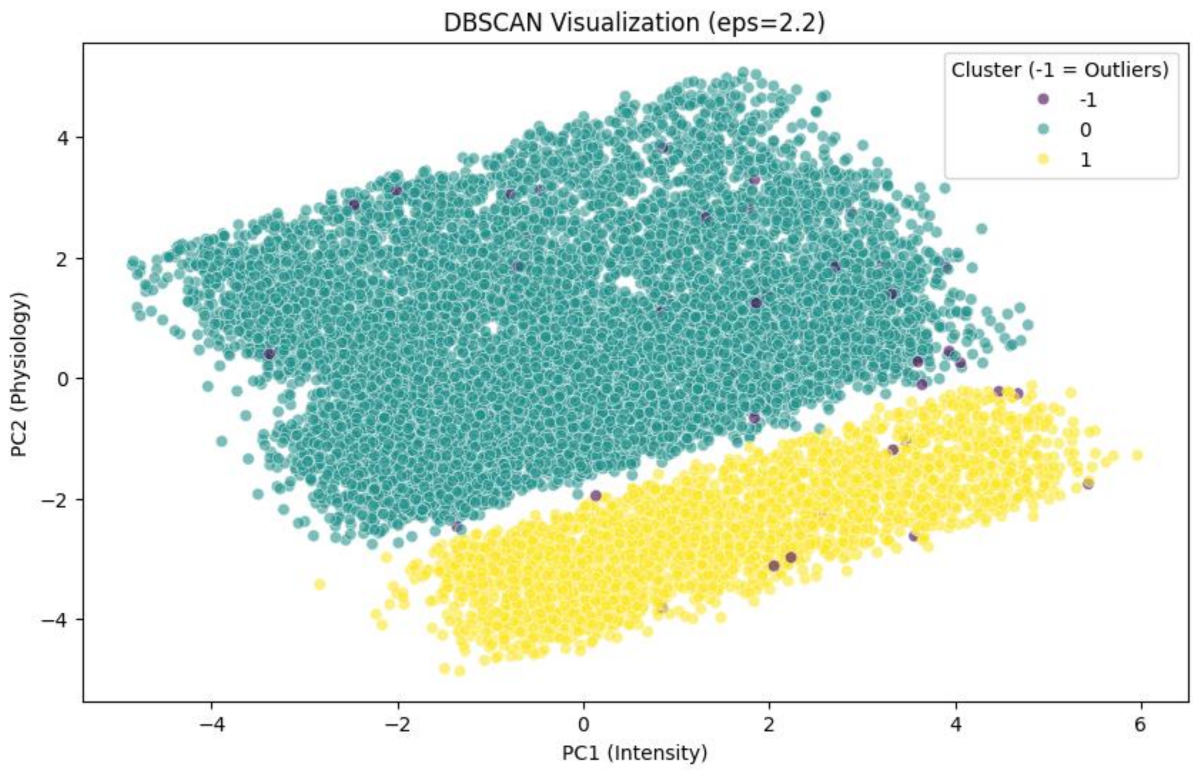
Step 3.1:
Euclidean Distance (Difference between users)
250
200
150
100
50
0
Users (Grouped for visibility)



Step 3.2: Hierarchical Silhouette Evaluation
Silhouette Score
0.16
0.14
0.12
0.10
0.08
Hierarchical
K-Means Best (0.1701)
2
3
4
5
6
7
8
9
10
Number of Clusters (K)


Hierarchical Clustering Visualization (K=2)
HC_Cluster
0
1
PC2
PC1
4
2
0
−2
−4
−4
−2
0
2
4
6


Step 4.1: K-Distance Graph (Finding eps)
Distance to 5th Nearest Neighbor (eps)
Users sorted by distance
3.0
2.5
2.0
1.5
1.0
0
2000
4000
6000
8000
10000
12000
14000



Inference :

1. **PCA Dimensionality Reduction:** The PCA model retained **47.22%** of data variance. This level of compression successfully translated 16 complex variables into a 2D "Fitness Universe" map without losing the core structure of the clusters.
2. **The Power of K=3:** Moving from 2 to 3 clusters in K-Means reduced internal cluster tension (Inertia) by **15.53%**. This transition moved the analysis from a simple binary split to a more nuanced three-tier persona system.
3. **Persona Identification:** The model successfully isolated three distinct groups based on experience:
4. **Cluster 0 (The Starters):** High volume, low experience.
5. **Cluster 1 (The Veterans):** High experience, elite metrics.
6. **Cluster 2 (The Mid-Tier):** Transitional users.
7. **Fundamental Binary Split:** The high Silhouette scores for **Hierarchical (0.1715)** and **DBSCAN (0.1718)** at K=2 confirm that the most basic division in fitness data is driven by **Heart Rate Intensity** (High-Intensity vs. Moderate-Intensity).
8. **Elite Outlier Detection:** DBSCAN identified **0.37%** of the population as outliers. These users represent "Elite Athletes" with the lowest fat percentages (**18.23%**) and longest workout durations (**1.05 hours**), providing the app with a way to handle edge-case performance data.
9. **Final Model Consensus:** The alignment between K-Means and PCA visuals validates that the calculated features (BMI, MET, HR Intensity) are the primary drivers of user segmentation.